

Temperature sensor

DETECT

INPUT

ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses



Plant propagator

DETECT
Measure the temperature

DECIDE
Too cold or too warm

DO
Control heat or ventilation

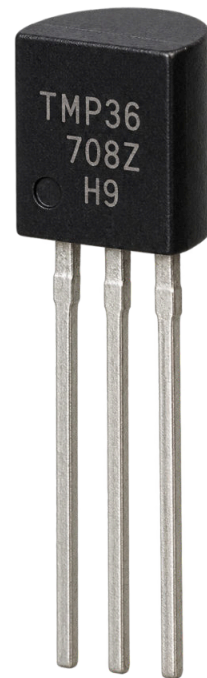


Soup drama meter

DETECT
Measure soup temperature

DECIDE
Safe or lava

DO
Move the drama pointer



Flat face

Use it to identify the pins

+Vs

Left pin

Vout

Centre pin

GND

Right pin

Code

● Raspberry Pi

```
from gpiozero import MCP3008

sensor = MCP3008(channel=0)

temperature = (sensor.voltage - 0.5) * 100
```

● Raspberry Pi Pico

```
from picozero import TemperatureSensor

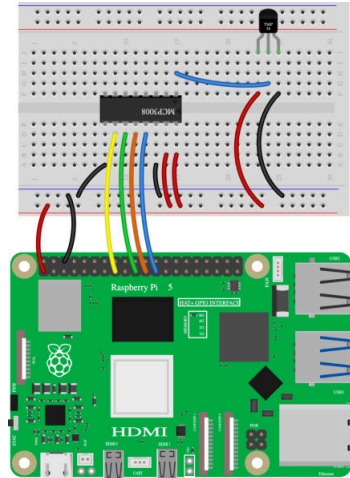
convert = lambda voltage: (voltage - 0.5) * 100
sensor = TemperatureSensor(26, conversion=convert)

temperature = sensor.temp
```

Read the TMP36 with its flat face towards you; the pins are +Vs, Vout, GND from left to right.

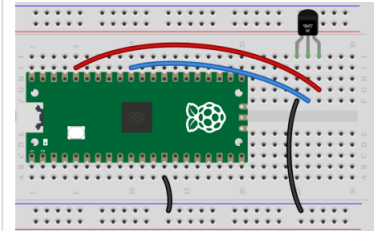
Wiring

RASPBERRY PI 5



TMP36 **Vout** → MCP3008 **CHO**; ADC SPI uses GPIO **22, 10, 9, 11**

PICO



TMP36 **+Vs** → 3V3 OUT, **Vout** → GP26, **GND** → GND